

Routers and Switches Example

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Routers and switches are both network connecting devices but both achieve their purpose in different ways. Routers have the responsibility of settling the shortest path for a packet to arrive at its intended destination. This device connects different networks. Switches have the main purpose of connecting different devices. Home routers are typically not designed to be the same as business-class devices due to the vast need for network support in a company vs. the few devices in a house. Business-class routers and switches vary in price depending on which specification is of a higher priority to the buyer. Though The Netgear Nighthawk X10 router, model, R9000, is sold by Amazon, Asus, Fruugo, and Netgear (*Athow, 2021*). It costs about \$299 depending on the seller, with a WiFi range of 2500 sqft, 5 Ethernet ports, and a Speed of 1734 Mbps, 7 Gigabit Ethernets, and can be bundled with Amazon Cloud Drive. Netgear Orbi Pro router (SXS50-100NAS), is sold by Amazon for \$199.99 and manufactured by Netgear. It has a WiFi range of 2500 sqft, flexible VLAN configuration, 4 SSID wireless networks, and has .6+4.8+2.4 Gbps WiFi speed (*Amazon, 2022*).

The NETGEAR 24-Port Gigabit Ethernet Unmanaged Switch (JGS524), is sold by Amazon for \$159 and is manufactured by Netgear. It has the switching Bandwidth of 48Gbps, 24 Gigabit Ethernet Ports, a forwarding rate of 10Mbps, is unmanaged, has fanless housing, suitable for rackmount or desktop setup. The TP-Link 16-Port Fast Ethernet Unmanaged Switch (TL-SF1016D) is sold by Amazon for \$39, with a Data Transfer Rate of 100Mbps, 16 ports, 8.5 Watts, and a 3.2Gbps Switching Capability(*Amazon, 2022*).

References

Amazon. (n.d.-a). *Amazon.com: NETGEAR 24-Port Gigabit Ethernet Unmanaged Switch (JGS524) - Desktop or Rackmount, and Limited Lifetime Protection : Electronics.*

<https://www.amazon.com/NETGEAR-Ethernet-Unmanaged-Protection->

[JGS524NA/dp/B0002CWPW2/ref=sr_1_3?crid=Z30ISDIEUNJB&keywords=netgear%2B24%2Bport%2Bgigabit&qid=1657422300&sprefix=netgear%2B24%2Bport%2Bgigabit%2Caps%2C92&sr=8-3&th=1](https://www.amazon.com/NETGEAR-Ethernet-Unmanaged-Protection-JGS524NA/dp/B0002CWPW2/ref=sr_1_3?crid=Z30ISDIEUNJB&keywords=netgear%2B24%2Bport%2Bgigabit&qid=1657422300&sprefix=netgear%2B24%2Bport%2Bgigabit%2Caps%2C92&sr=8-3&th=1)

Amazon. (n.d.-b). *Amazon.com: NETGEAR Nighthawk X10 Smart WiFi Router (R9000) - AD7200 Wireless Speed (up to 7200 Mbps) for 60Ghz WiFi Devices | Up to 2500 sq ft Coverage | 6 x 1G Ethernet, 1 x 10G SFP+, and 2 USB ports : Electronics.*

<https://www.amazon.com/dp/B01M12RE4A?tag=georiot-us-default->

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Amazon. (n.d.-c). *Amazon.com: NETGEAR Orbi Pro WiFi 6 Tri-Band Mesh Add-on Satellite (SXS50) for Business or Home, Coverage up to 2,500 sq. ft, 75 Devices, AX5400 802.11AX (up to 5.4Gbps), Requires Orbi Pro WiFi 6 Router : Everything Else.*

<https://www.amazon.com/NETGEAR-Tri-Band-Satellite-Extenders->

[Business/dp/B09PX728P8/ref=sr_1_6?gclid=Cj0KCQjwzqSWBhDPAIsAK38LY83ZjtCwK_ACq7DIRNcQE1j6a9pAWbObmWoUm98l3EO0U7d00yatgwaApSDEALw_wcB&hvadid=557373286068&hvdev=c&hvlocphy=9021581&hvnetw=g&hvqmt=e&hvrant=7475120972333968102&hvtargid=kwd-](https://www.amazon.com/NETGEAR-Tri-Band-Satellite-Extenders-Business/dp/B09PX728P8/ref=sr_1_6?gclid=Cj0KCQjwzqSWBhDPAIsAK38LY83ZjtCwK_ACq7DIRNcQE1j6a9pAWbObmWoUm98l3EO0U7d00yatgwaApSDEALw_wcB&hvadid=557373286068&hvdev=c&hvlocphy=9021581&hvnetw=g&hvqmt=e&hvrant=7475120972333968102&hvtargid=kwd-)

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Amazon. (n.d.-d). *Amazon.com: TP-Link 16 Port 10/100Mbps Fast Ethernet Switch | Desktop or Wall-Mounting | Plastic Case Ethernet Splitter | Unshielded Network Switch | Plug and Play | Fanless Quiet | Unmanaged (TL-SF1016D) : Electronics.*

<https://www.amazon.com/TP-Link-16-Port-Ethernet-Unmanaged-TL-SF1016D/dp/B003CFATTM?th=1>

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GeeksforGeeks. (2021, July 27). *Difference between Router and Switch.* Retrieved July 11, 2022, from <https://www.geeksforgeeks.org/difference-between-router-and-switch/>